

hw0317_3_23

2026-03-22

```
df <- read.csv("/Users/chenjunqi/Desktop/114-2/計量經濟/poe5csv/collegetown.csv")
```

```
df$sqft2 <- df$sqft^2
model <- lm(price ~ sqft2, data = df)
summary(model)
```

```
##
## Call:
## lm(formula = price ~ sqft2, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -383.67  -48.39   -7.50   38.75  469.70
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 93.565854   6.072226  15.41  <2e-16 ***
## sqft2        0.184519   0.005256  35.11  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 92.08 on 498 degrees of freedom
## Multiple R-squared:  0.7122, Adjusted R-squared:  0.7117
## F-statistic: 1233 on 1 and 498 DF, p-value: < 2.2e-16
```

```
coef_est <- coef(summary(model))
alpha2_hat <- coef_est["sqft2", "Estimate"]
se_alpha2 <- coef_est["sqft2", "Std. Error"]
```

#a

```
# 2000 square feet = 20 個「百平方英尺」
x0 <- 20

# marginal effect = dE(PRICE|SQFT)/dSQFT = 2 * alpha2 * SQFT
# 在 SQFT = 20 時, marginal effect = 40 * alpha2
marginal_hat <- 2 * alpha2_hat * x0

# H0: marginal effect <= 13
# H1: marginal effect > 13
# 因為 PRICE 單位是千美元, 所以 $13000 = 13
null_value <- 13

# marginal effect 的標準誤
se_marginal <- 2 * x0 * se_alpha2
```

```
# t 統計量
t_stat <- (marginal_hat - null_value) / se_marginal

# 自由度
dfree <- df.residual(model)

# 右尾檢定 p-value
p_value <- 1 - pt(t_stat, df = dfree)

# 5% 顯著水準右尾臨界值
t_crit <- qt(0.95, df = dfree)
```

```
# 輸出結果
cat("alpha2_hat =", alpha2_hat, "\n")
```

```
## alpha2_hat = 0.184519
```

```
cat("SE(alpha2_hat) =", se_alpha2, "\n")
```

```
## SE(alpha2_hat) = 0.005255801
```

```
cat("Estimated marginal effect at SQFT = 20 =", marginal_hat, "\n")
```

```
## Estimated marginal effect at SQFT = 20 = 7.38076
```

```
cat("SE of marginal effect =", se_marginal, "\n")
```

```
## SE of marginal effect = 0.210232
```

```
cat("t statistic =", t_stat, "\n")
```

```
## t statistic = -26.72875
```

```
cat("Critical value =", t_crit, "\n")
```

```
## Critical value = 1.647919
```

```
cat("p-value =", p_value, "\n")
```

```
## p-value = 1
```

```
# 結論
if (t_stat > t_crit) {
  cat("Reject H0: the marginal effect is greater than $13,000.\n")
} else {
  cat("Do not reject H0: insufficient evidence that the marginal effect is greater than $13,000.\n")
}
```

```
## Do not reject H0: insufficient evidence that the marginal effect is greater than $13,000.
```

$$\text{Model: } PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$$

The conditional expectation function is

$$E(PRICE \mid SQFT) = \alpha_1 + \alpha_2 SQFT^2.$$

Hence, the marginal effect of house size on expected price is

$$\frac{dE(PRICE \mid SQFT)}{dSQFT} = 2\alpha_2 SQFT.$$

Since $SQFT$ is measured in hundreds of square feet, a 2000 square foot house corresponds to

$$SQFT = 20.$$

Therefore, the marginal effect at $SQFT = 20$ is

$$2\alpha_2(20) = 40\alpha_2.$$

Because $PRICE$ is measured in thousands of dollars, \$13,000 corresponds to 13.

Thus, the hypotheses are

$$H_0 : 40\alpha_2 \leq 13 \quad \text{versus} \quad H_1 : 40\alpha_2 > 13.$$

From the regression results,

$$\hat{\alpha}_2 = 0.18452, \quad se(\hat{\alpha}_2) = 0.005256.$$

Thus, the estimated marginal effect at $SQFT = 20$ is

$$40\hat{\alpha}_2 = 40(0.18452) = 7.381.$$

The standard error of $40\hat{\alpha}_2$ is

$$se(40\hat{\alpha}_2) = 40 \cdot se(\hat{\alpha}_2) = 40(0.005256) = 0.21024.$$

So the test statistic is

$$t = \frac{40\hat{\alpha}_2 - 13}{40 se(\hat{\alpha}_2)} = \frac{7.381 - 13}{0.21024} \approx -26.73.$$

With $n = 500$, the degrees of freedom are

$$df = 500 - 2 = 498.$$

At the 5% significance level, this is a one-sided test, so the rejection region is

$$t > t_{0.95, 498} \approx 1.648.$$

The p-value is

$$p\text{-value} = P(T_{498} > -26.73) \approx 1.000.$$

Since

$$-26.73 < 1.648,$$

we do not reject H_0 . There is insufficient evidence at the 5% significance level to conclude that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is greater than \$13,000. The data do not support the claim.

#b

```
# b 小題: 4000 square feet = 40 個「百平方英尺」
x1 <- 40

# marginal effect = dE(PRICE|SQFT)/dSQFT = 2 * alpha2 * SQFT
marginal_hat <- 2 * alpha2_hat * x1

# H0: marginal effect <= 13
# H1: marginal effect > 13
null_value <- 13

# marginal effect 的標準誤
se_marginal <- 2 * x1 * se_alpha2

# t statistic
t_stat <- (marginal_hat - null_value) / se_marginal

# 自由度
dfree <- df.residual(model)

# 右尾檢定 p-value
p_value <- 1 - pt(t_stat, df = dfree)

# 5% 顯著水準右尾臨界值
t_crit <- qt(0.95, df = dfree)

# 輸出結果
cat("Estimated marginal effect at SQFT = 40:", marginal_hat, "\n")
```

```
## Estimated marginal effect at SQFT = 40: 14.76152
```

```
cat("Standard error:", se_marginal, "\n")
```

```
## Standard error: 0.4204641
```

```
cat("t statistic:", t_stat, "\n")
```

```
## t statistic: 4.189467
```

```
cat("Critical value:", t_crit, "\n")
```

```
## Critical value: 1.647919
```

```
cat("p-value:", p_value, "\n")
```

```
## p-value: 1.65395e-05
```

```
# 結論
if (t_stat > t_crit) {
  cat("Reject H0: the marginal effect is greater than $13,000.\n")
} else {
  cat("Do not reject H0: insufficient evidence that the marginal effect is greater than $13,000.\n")
}
```

```
## Reject H0: the marginal effect is greater than $13,000.
```

Using the quadratic regression model $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$,

$$E(PRICE \mid SQFT) = \alpha_1 + \alpha_2 SQFT^2.$$

The marginal effect is

$$\frac{dE(PRICE \mid SQFT)}{dSQFT} = 2\alpha_2 SQFT.$$

Since $SQFT$ is measured in hundreds of square feet, a 4000 square foot house corresponds to

$$SQFT = 40.$$

Thus, the marginal effect at $SQFT = 40$ is

$$2\alpha_2(40) = 80\alpha_2.$$

Because $PRICE$ is measured in thousands of dollars, \$13,000 corresponds to 13. Therefore, the hypotheses are

$$H_0 : 80\alpha_2 \leq 13 \quad \text{versus} \quad H_1 : 80\alpha_2 > 13.$$

From part (a),

$$\hat{\alpha}_2 = 0.18452, \quad se(\hat{\alpha}_2) = 0.005256.$$

Hence, the estimated marginal effect at $SQFT = 40$ is

$$80\hat{\alpha}_2 = 80(0.18452) = 14.7616.$$

Its standard error is

$$se(80\hat{\alpha}_2) = 80(0.005256) = 0.42048.$$

So the test statistic is

$$t = \frac{80\hat{\alpha}_2 - 13}{80 se(\hat{\alpha}_2)} = \frac{14.7616 - 13}{0.42048} \approx 4.1895.$$

With $n = 500$, the degrees of freedom are

$$df = 500 - 2 = 498.$$

At the 5% significance level, the rejection region is

$$t > t_{0.95, 498} \approx 1.648.$$

The p-value is

$$p\text{-value} = P(T_{498} > 4.19) \approx 1.7 \times 10^{-5}.$$

Since

$$4.1895 > 1.648,$$

we reject H_0 .

At the 5% significance level, the data support the claim that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is greater than \$13,000.

#c

```
# 2000 square feet = 20 個「百平方英尺」
newdata <- data.frame(sqft2 = 20^2)

# 估計期望價格與 95% 信賴區間
pred_ci <- predict(model, newdata = newdata, interval = "confidence", level = 0.95)

# 印出結果
print(pred_ci)
```

```
##          fit      lwr      upr
## 1 167.3735 158.0481 176.6988
```

Using the quadratic regression model $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$,

$$E(PRICE \mid SQFT) = \alpha_1 + \alpha_2 SQFT^2.$$

Since $SQFT$ is measured in hundreds of square feet, a 2000 square foot house corresponds to

$$SQFT = 20.$$

Using the estimated regression equation,

$$E(PRICE \mid \widehat{SQFT} = 20) = \hat{\alpha}_1 + \hat{\alpha}_2(20)^2 = 93.5659 + 0.184519(400) = 167.3735.$$

Thus, the estimated expected price is

$$167.37$$

in thousand-dollar units, i.e. approximately \$167,373.

A 95% confidence interval for the expected price is

$$(158.05, 176.70)$$

in thousand-dollar units.

Therefore, we estimate that the average price of a 2000 square foot house is about \$167,373, and we are 95% confident that the true expected price lies between \$158,048 and \$176,699.

#d

```
# 找出 SQFT = 20 的房子
subset_20 <- subset(df, sqft == 20)
```

```
# 看有幾間
nrow(subset_20)
```

```
## [1] 3
```

```
# 顯示這些房子的價格
subset_20$price
```

```
## [1] 138 169 183
```

```
# 計算樣本平均售價
mean_price_20 <- mean(subset_20$price)
mean_price_20
```

```
## [1] 163.3333
```

Houses in the sample with $SQFT = 20$ have selling prices

$$138, 169, 183$$

(in thousand-dollar units). Therefore, the sample mean selling price is

$$PRICE = \frac{138 + 169 + 183}{3} = 163.33.$$

Thus, the sample average selling price for houses with $SQFT = 20$ is

$$163.33$$

in thousand-dollar units. In part (c), the estimated expected price for a house with $SQFT = 20$ was

$$E(PRICE \mid \widehat{SQFT} = 20) = 167.37$$

with a 95% confidence interval

(158.05, 176.70).

Since

$$163.33 \in (158.05, 176.70),$$

the sample average selling price is compatible with the result in part (c). The observed sample mean is close to the estimated expected price and lies within the 95% confidence interval, so there is no contradiction between the sample evidence and the model-based estimate.